

E.3 / PAPER 2 / QUESTION 8

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8. Magnesium-27 nuclei ($^{27}_{12}\text{Mg}$) decay by beta-minus (β^-) decay to form nuclei of aluminium-27 (Al).

- (a) Show, using the data, that the energy released in the decay of one magnesium-27 nucleus is about 2.62 MeV.

Mass of aluminium-27 atom = 26.98153 u

Mass of magnesium-27 atom = 26.98434 u

The unified atomic mass unit is 931.5 MeV c^{-2} . [1]

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- (b) A Magnesium-27 nucleus can decay by one of two routes:

Route 1: 70 % of the beta particles are emitted with a maximum kinetic energy of 1.76656 MeV, accompanied by a gamma photon of energy 0.84376 MeV.

Route 2: 30 % of the beta particles have a maximum kinetic energy of 1.59587 MeV with a gamma photon of energy 1.01445 MeV.

The final state of the aluminium-27 nucleus is the same for both routes.

- (i) State the conclusion that can be drawn from the existence of these two routes. [1]

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- (ii) Calculate the difference between the magnitudes of the total energy transfers in parts (a) and (b). [1]

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- (iii) Explain how the difference in part (b)(ii) arises. [1]

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- (c) Small amounts of magnesium in a material can be detected by firing neutrons at magnesium-26 nuclei. This process is known as irradiation.

Magnesium-27 is formed because of irradiation. The products of the beta-particle emission are observed as the magnesium-27 decays to aluminium-27.

- (i) The smallest mass of magnesium that can be detected with this technique is 1.1×10^{-8} kg.

Show that the smallest number of magnesium atoms that can be detected with this technique is about 10^{17} .

[2]

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- (ii) A sample of glass is irradiated with neutrons so that all the magnesium atoms become magnesium-27. The sample contains 9.50×10^{15} magnesium atoms.

The decay constant of magnesium-27 is $1.22 \times 10^{-3} \text{ s}^{-1}$.

Determine the number of aluminium atoms that form in 10.0 minutes after the irradiation ends.

[3]

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- (iii) Estimate, in W, the average rate at which energy is transferred by the decay of magnesium-27 during the 10.0 minutes after the irradiation ends.

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References:

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